

REESE AFB, TX, USA

WMO: 722675

Lat: 33.600N

Lon: 102.050W

Elev: 3337

StdP: 13.01

Time Zone: -6.00 (NAC)

Period: 86-97

WBAN: 23021

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.7	19.4	0.3	6.3	23.3	4.2	7.7	32.2	31.7	54.4	28.2	50.6	8.2	20	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	101.0	67.0	97.8	67.3	95.1	67.2	73.2	87.2	72.0	86.7	71.0	86.2	13.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	122.9	78.7	67.9	116.4	77.6	66.6	110.9	76.9	39.1	86.9	37.8	86.6	36.9	86.1	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.4	24.1	20.6	DB	7.8	105.4	5.3	4.8	4.0	108.9	1.0	111.7	-2.0	114.4	-5.8	117.9	
			WB	6.3	75.6	4.9	2.5	2.8	77.4	-0.1	78.8	-2.9	80.2	-6.5	82.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	41.1	45.5	52.6	62.2	70.7	78.1	80.4	78.2	71.3	62.1	50.1	42.3	(6)
	DBStd	16.11	8.70	10.23	9.72	8.43	7.78	5.76	4.68	4.96	6.60	7.82	9.26	9.61	(7)
	HDD50	965	291	182	87	11	1	0	0	0	1	12	112	268	(8)
	HDD65	3182	741	546	392	147	35	1	0	2	22	142	450	704	(9)
	CDD50	5091	16	57	168	376	643	843	943	874	639	388	115	29	(10)
	CDD65	1831	0	1	7	62	212	394	478	411	210	53	3	0	(11)
	CDH74	22615	11	65	363	1168	2633	4700	5782	4561	2223	967	122	20	(12)
	CDH80	11290	0	11	95	453	1296	2554	3197	2354	952	356	21	1	(13)
(14) Wind	WSAvg	9.8	9.6	10.2	11.5	11.6	11.9	10.3	9.0	7.6	7.7	8.5	10.4	9.5	(14)
(15) Precipitation	PrecAvg	18.6	0.8	0.7	1.2	1.4	2.5	2.7	2.2	2.0	2.0	1.4	0.9	0.7	(15)
	PrecMax	35.3	1.8	2.2	5.1	4.7	5.0	6.2	5.5	4.0	5.8	3.9	5.6	2.7	(16)
	PrecMin	10.4	0.1	0.0	0.0	0.0	0.4	0.2	0.1	0.4	0.0	0.0	0.0	0.0	(17)
	PrecStd	5.4	0.6	0.7	1.1	1.2	1.5	1.9	1.3	1.1	1.8	1.2	1.2	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	81.2	87.1	94.6	103.5	105.3	104.8	100.0	97.3	91.5	82.6	76.9	(19)
		MCWB	50.1	52.9	56.0	60.6	62.9	67.5	68.2	69.4	66.2	60.8	55.0	51.4	(20)
	2%	DB	68.4	75.1	82.7	88.9	96.8	100.0	100.9	97.2	92.5	88.0	77.4	71.3	(21)
		MCWB	47.8	51.6	54.0	57.5	62.9	66.7	68.3	69.0	66.1	60.1	54.1	49.6	(22)
	5%	DB	63.4	69.8	78.2	85.4	91.7	96.4	97.6	94.8	89.6	84.2	72.8	65.1	(23)
		MCWB	45.8	49.1	53.2	56.5	62.3	66.4	68.5	68.8	66.0	59.2	52.7	47.3	(24)
	10%	DB	58.7	64.7	72.8	81.4	87.3	93.0	94.4	91.9	85.8	79.7	67.6	59.9	(25)
		MCWB	43.1	46.8	51.6	55.8	62.5	67.4	68.3	68.8	66.3	59.1	50.7	44.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	55.1	60.5	66.1	70.7	74.7	75.0	74.6	73.2	66.5	60.3	54.3	(27)
		MCDB	65.6	72.7	71.5	82.2	85.2	86.6	89.0	88.6	86.0	80.6	70.1	68.4	(28)
	2%	WB	49.3	52.8	57.8	62.5	68.7	72.7	72.9	72.9	70.8	63.4	57.9	51.2	(29)
		MCDB	64.4	70.8	74.2	78.3	83.5	86.0	87.7	87.6	83.7	79.4	68.5	65.6	(30)
	5%	WB	46.8	50.7	55.3	60.4	66.9	71.4	71.4	71.5	69.3	62.0	55.4	48.3	(31)
		MCDB	61.9	67.2	72.6	76.5	81.4	84.9	88.0	86.9	82.0	76.7	68.8	62.9	(32)
	10%	WB	44.2	48.2	52.9	58.3	65.5	70.2	70.5	70.5	68.1	60.2	52.6	45.7	(33)
		MCDB	57.1	62.4	69.9	74.9	80.1	85.0	86.9	86.2	80.7	75.1	65.6	58.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	26.7	27.2	30.3	30.0	26.5	25.5	24.6	23.1	24.1	29.1	27.7	27.1	(35)
		MCDBR	36.5	37.3	38.8	36.8	33.1	30.7	30.0	28.6	30.0	35.7	34.5	37.4	(36)
	5% WB	MCWBR	20.5	18.9	18.8	15.6	12.4	9.3	9.0	9.3	11.0	14.4	17.5	21.0	(37)
		MCWBR	33.7	33.4	33.7	29.9	26.4	24.9	24.0	23.9	22.2	29.6	28.5	33.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.260	0.271	0.296	0.329	0.341	0.381	0.386	0.373	0.347	0.317	0.287	0.268	(40)
		taud	2.578	2.556	2.494	2.393	2.422	2.355	2.388	2.389	2.438	2.497	2.513	2.548	(41)
	Ebn at Noon		309	315	312	304	299	285	283	286	291	293	294	299	(42)
		Edh at Noon	24	28	32	38	37	39	38	37	34	30	26	24	(43)
(44) All-Sky Solar Radiation	RadAvg		1067	1321	1718	2113	2270	2405	2300	2117	1752	1397	1112	927	(44)
	RadStd		101	120	124	143	171	134	111	92	121	154	113	71	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+131

Nomenclature: See separate page